Assignment 2 :

1\. Two numbers MN_H and KL_H are stored in memory locations 2050_H and 2051_H respectively. Write a program to assemble them as NK_H and LM_H. Store them in 2052_H and 2053_H respectively.

Solution:

Instructions start from $E000_H$

Data: ⇒ MN_H ⇒ $[E100_H]$

KL_H → $[E101_H]$

Algorithm:

$L \leftarrow 00_H$

$H \leftarrow E1_H$

$A \leftarrow [HL]$

Rotate Left 4 times ($MN_H \rightarrow NM_H$)

$B \leftarrow A$

$HL \leftarrow HL + 1$

$A \leftarrow [HL]$

Rotate Left 4 times ($KL_H \rightarrow LK_H$)

$C \leftarrow A$

Extract L from LK_H

$D \leftarrow A$

$A \leftarrow B$

Extract M from NM_H

ADD $L0_H$ and $0M_H$

$[E103_H] \leftarrow LM_H$

Repeat this for getting NK_H and store in $[E102_H]$

Code:

opcode	Hexcode	Description
LXI H, E100H	21 00 E1	Load HL Immediate.
MOV A, M	7E	Move from [HL] to A
RLC	07	Rotate accumulator
RLC	07	without carry
RLC	07	
RLC	07	
MOV B, A	47	Move value of A to B
INX H	23	increment HL reg pair
MOV A, M	7E	Move from [HL] to A
RLC	07	Rotate accumulator
RLC	07	left without
RLC	07	carry.
RLC	07	
MOV C, A	4F	Move value of C to A
ANI F0H	E6 F0	And immediate accumulator
MOV D, A	57	
MOV A, B	78	
ANI 0FH	E6 0F	And immediate accumulator.
ADD D	82	Add accumulator with D.
STA E103H	32 03 E1	Store accumulator to m/m
MOV A, B	78	
ANI F0H	E6 F0	
MOV D, A	57	
MOV A, C	79	
ANI 0FH	E6 0F	
ADD D	82	
STA E102H	32 02 E1	
HLT	76	

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Question - 2

Two numbers A and B are stored in 2050H and 2051H respectively. Write a program to perform A×B and store result in 2052H and 2053H.

Solution:

Instructions start from: E000H

Data: A => [E100H]

B => [E101H]

Result: [E102H] => LSB

[E103H] => MSB

Algorithm:

```
L ← [E100H]
H ← [E101H]
DE ← HL        (Exchange DE and HL)
C ← D
D ← 00H        ( C = [E101H] = BH ) (E = [E100H] = AH)
H ← 00H
L ← 00H
Loop:   HL = HL + DE   } Add AH for BH times
        C ← C - 1      }
        If (C != 0) goto LOOP
[E102] ← L
[E103] ← H
END
```

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Code:

Opcode	Hexcode	Description
LHLD E100H	2A 00 E1	Load HL direct from m/m
XCHG	EB	Exchange DE and HL
MOV C, D	4A	move value from D to C
MVI D, 00H	16 00	move immediate 00H to D
LXI H, 0000H	21 00 00	Load immediate 0000H to HL
LOOP: DAD D	19	Double add HL and DE
DCR C	0D	decrement C
JNZ LOOP	C2 0A E0	If C is not 0, jump to LOOP
SHLD E102H	22 02 E1	Store HL direct to m/m
HLT	76	Stop.

Result:

Data: A = [E100H] = 9AH

B = [E101H] = 4CH

[E102H] = B8H

[E103H] = 2DH

$\therefore (9A)_{16} \times (4C)_{16} = (2D\ B8)_{16}$

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Question 3:

N numbers are stored in consecutive m/m locations starting from 2050H. The value of N is stored in 204FH.

⇒ Data: N => E200 H

numbers => E100 H onwards.

Instructions from E000 H

ii) find maximum among N numbers.

Algorithm:

```
A ← [E200H]
C ← A
D ← 0
H ← E1H
L ← 00H
loop:   A ← [HL]
        if (A <D)
            goto Next
        D ← [HL]
Next:   HL ← HL+1
        C ← C-1
        if (C != 0)
            goto loop
    A ← D
    [E300] ← A
    END
```

} if A > D, store A in D

} finally D has maximum value.

Code:

Opcode	hexcode	Description
LDA E200H	3A 00 E2	Load accumulator direct
MOV C, A	4F	move value of C to A
MVI D, 00H	16 00	
LXI H, E100H	21 00 E1	load immediate to HL
LOOP: MOV A, M	7E	
CMP D	BA	Compare accumulator with D
JC NEXT	DA 0E E0	Jump on carry
JZ NEXT	CA 0E E0	Jump on zero
MOV D, M	56	
NEXT: INX H	23	increase value of HL
DCR C	0D	
JNZ LOOP	C2 09 E0	
MOV A, D	7A	
STA E300H	32 00 E3	
HLT	76	

Result:

numbers: A9, 91, 04, 12, C4

output: C4

(i) Find minimum among N numbers.

Algorithm:

A ← [E200H]

C ← A

D ← FFH

H ← E1H   L ← 00H

loop: A ← [HL]

 if (A >= D)

  goto Next

 D ← [HL]

Next: HL ← HL+1   C ← C-1

 if (C != 0)

  goto loop

A ← D

[E300] ← A

END.

Code:

Opcode	Hexcode	Description
LDA E200H	3A 00 E2	Load Accumulator direct
MOV C, A	4F	
MVI D, FFH	16 00	
LXI H, E100H	21 00 E1	
LOOP: MOV A, M	7E	
CMP D	BA	
JNC NEXT	DA 0D E0	Jump on no carry to Next

Opcode	Hexcode	Description
MOV D, M	56	
NEXT: INX H	23	
DCR C	0D	
JNZ LOOP	C2 09 E0	
MOV A, D	7A	
STA E300H	32 00 E3	
HLT	76	

Result: numbers: A9, 91, 04, 12, C4

Output: 04

(iii) Sort N numbers in Ascending Number

Algorithm:

loop: H ← E2H L ← 00H

D ← 00H

C ← [HL]

C ← C-1 HL ← HL+1

check: A ← [HL]

HL ← HL+1

if (A > [HL]) swap, D ← 01H } Value of D checks if swap has occured.

else goto Next

Next: C ← C-1

if (C! = 0) goto check

A ← D

if (D == 01) goto loop } swap occured in last iteration

} if no swap, end.

End

Code:

opcode		Hexcode	Description
Loop:	LXI H, E200H	21 00 E2	load HL immediate
	MVI D, 00H	16 00	move 00H to D
	MOV C, M	4E	move value of m/m to C
	DCR C	0D	decrement value of C
	INX H	23	increment HL
CHECK:	MOV A, M	7E	
	INX H	23	
	CMP M	BE	Compare accumulator with HL
	JC NEXT	DA 18 E0	Jump on carry
	JZ NEXT	CA 18 E0	Jump on zero
	MOV B, M	46	
	MOV M, A	77	
	DCX H	2B	decrement HL rp.
	MOV M, B	70	
	INX H	23	
	MVI D, 01H	16 01	
NEXT:	DCR C	0D	
	JNZ CHECK	C2 08 E0	Jump on no zero.
	MOV A, D	7A	
	CPI 01H	FE 01	
	JZ LOOP	CA 00 E0	
	HLT	76	

Result:

Data: A9, 1C, 04, 91, 12

Result: 04, 12, 1C, 91, A9

(iv) Sort N numbers in descending order.

Algorithm:

```
loop:   H ← E2H    L ← 00H
        D ← 00H
        C ← [HL]
        C ← C-1
        HL ← HL+1
check:  A ← [HL]
        HL ← HL+1
        if (A < [HL])
            swap [HL] and [HL-1]
            D ← 01H
Next:   C ← C-1
        if (C != 0)
            goto check
        A ← D
        if (D == 01)     } D = 01 => Exchange occured in
            goto loop      last iteration. If D = 00H, no
                           exchange in last iteration => data
END                        is sorted.
```

* The outer loop runs till exchanges occur in the inner loop.

Code:

Opcode	hexcode	Description
LOOP: LXI H, E0FFH	21 FF E0	
MVI D, 00H	16 00	
MOV C, M	4E	
DCR C	0D	
INX H	23	
CHECK: MOV A,M	7E	
INX H	23	
CMP M	BE	
JNC NEXT	D2 18 E0	
JZ NEXT	CA 18 E0	
MOV B, M	46	
MOV M, A	77	
DCX H	2B	
MOV M, B	70	
INX H	23	
MVI D, 01H	16 01	
NEXT: DCR C	0D	
JNZ CHECK	C2 08 E0	
MOV A, D	7A	
CPI 01H	FE 01	
JZ LOOP	CA 00 E0	
HLT	76	

Result: Data: A9, 1C, 04, 91, 12

result: A9, 91, 1C, 12, 04

Question 5:

N numbers are stored in m/m locations starting from 2050H. The value of N is stored in 204F. Write a program to test whether a number stored in 204E is present in the list. If present, store the position at 2040H. Otherwise store FFH.

Data: N: E200H

numbers: E100H onwords

number to be found: E201H

Instructions start from E000H.

~~opcod~~ Algorithm:

```
    A ← [E200]        (A ← N)
    C ← A
    D ← 01H           (position)
    A ← [E201]
    MOV B, A
    HL ← E100H
LOOP:  A ← [HL]
       if (A == B) goto Result.
       HL ← HL+1
       C ← C-1
       if (C != 0) goto LOOP
     A ← FFH
     goto FINAL
Result:  A ← D
FINAL:   [E300] ← A
HLT
```

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Code:

opcode	Hexcode	Description
LDA E200H	3A 00 E2	load accumulator direct.
MOV C, A	4F	move value from C to A
MVI D, 01H	16 01	move 01H to D
LDA E201H	3A 01 E2	
MOV B, A	47	
LXI H, E100H	21 00 E1	Load HL pair immediate
LOOP: MOV A, M	7E	
CMP B	B8	
JZ Result	CA 1D E0	
INX H	23	
INR D	14	
DCR C	0D	
JNZ LOOP	C2 0D E0	
MVI A, FFH	3E FF	
JMP FINAL	C3 1E E0	unconditional jump
Result: MOV A, D	7A	
FINAL: STA E300H	32 00 E3	Store accumulator to m/m
HLT	76	

Result:

Data: A9, 91, 1C, 12, 04

numbers to be found: 1C result -> 03

24 result -> FF

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Question 4:

N numbers are stored in consecutive m/m locations starting from 2050H. The value of N is stored in 204FH. Write a program to copy the even and odd numbers starting from 2100H and 2200H respectively. Store the total number of even and odd numbers in 2300H and 2201H respectively.

Data: numbers: E100H onwards.
N: E200H
Result: Even: E300 onwards no of even : E201H
Odd: E350 onwards no of odd : E202H

Algorithm:

Part 1: Load N to A.
C ← A
B ← 00H
Load address of even numbers to DE
Load address of first number to HL
LOOP1: MOV A, M
HL ← HL + 1
if (LSB of A == 0) goto EVEN
C ← C - 1
if (C != 0) goto LOOP1
goto FIN1
EVEN: Store Accumulator
DE ← DE + 1
B ← B + 1
C ← C - 1
if (C != 0) goto LOOP1
Store B.

(*) Repeat the same loop for odd numbers after changing the check condition to LSB == 1

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Code:

opcode	hexcode	Description
PART1: LDA E200H	3A 00 E2	PART1 calculates no of even numbers and stores them.
MOV C, A	4F	
MVI B, 00H	06 00	
LXI D, E300H	11 00 E3	
LXI H, E100H	21 00 E1	
LOOP1: MOV A, M	7E	
INX H	23	
RRC	0F	Checks LSB for odd or even.
JNC EVEN	D2 19 E0	
DCR C	0D	
JNZ LOOP1	C2 0C E0	
JMP FIN1	C3 21 E0	
EVEN: RLC	07	
STAX D	12	Store even
INX D	13	
INR B	04	
DCR C	0D	
JNZ LOOP1	C2 0C E0	
FIN1: MOV A, B	78	
STA E201H	32 01 E2	Store no of even.
PART2: LDA E200H	3A 00 E2	PART 2 calculates no of odd numbers and stores them. The code is similar to PART 1.
MOV C, A	4F	
MVI B, 00H	06 00	
LXI D, E350H	11 50 E3	
LXI H, E100H	21 00 E1	
LOOP2: MOV A, M	7E	
RRC	0F	

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opcode	hexcode	Description
JC ODD	DA 41 E0	LSB check condition is opposite of PART 1
INX H	23	
DCR C	0D	
JNZ LOOP2	C2 34 E0	
JMP FIN2	C3 4A E0	
ODD: RLC	07	
STAX D	12	Store odd number
INX D	13	
INX M	23	
INR B	04	
DCR C	0D	
JNZ LOOP2	C2 34 E0	
FIN2: MOV A, B	78	
STA E202 H	32 02 E2	Storing no of odd
HLT	76	

Result: Data: N = 5

Array = A1, 29, E0, C4, 11

Even : E0 C4 no of even = 2

Odd : A1, 29, 11 no of odd = 3

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